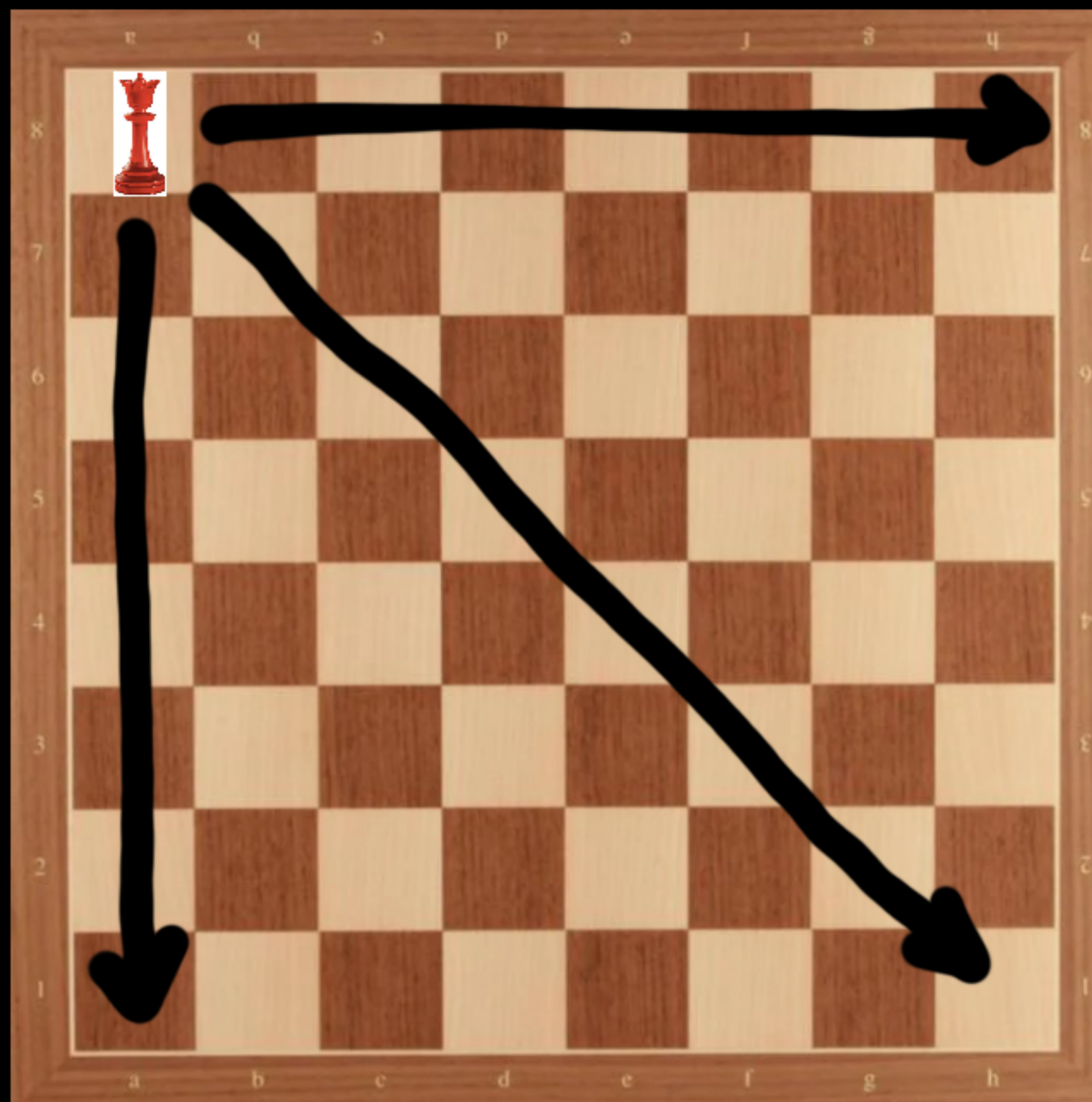


① 8-Queens Problem

8x8 chess board
Queens attack in one move
across a row
down a column
along the diagonal

Can you place 8 queens on
the board such that they
cannot attack each other?



②

Yes!

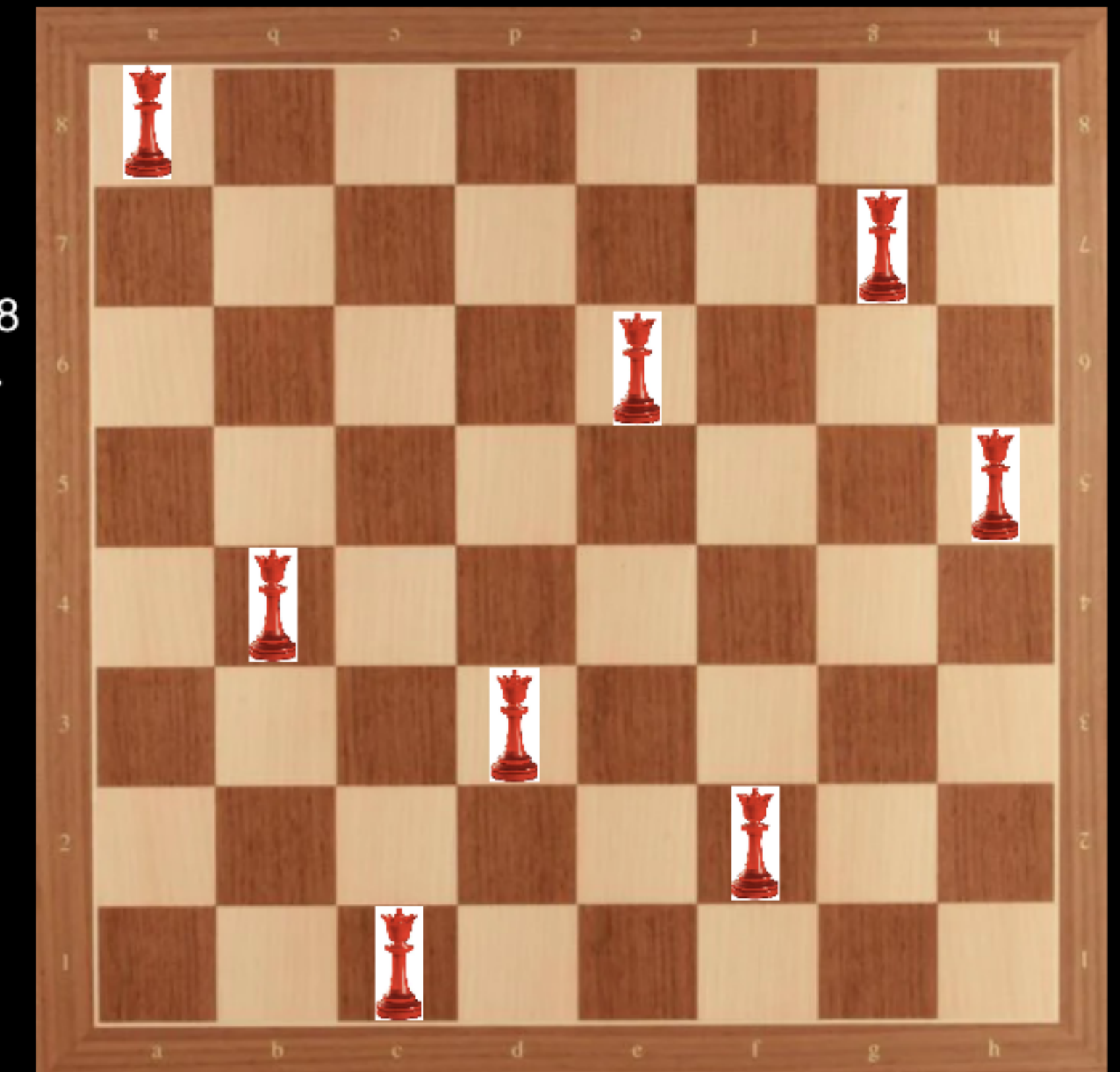
Here is one of the solutions.

There are $C(8, 64) = 4,426,165,368$
possible arrangements of queens.

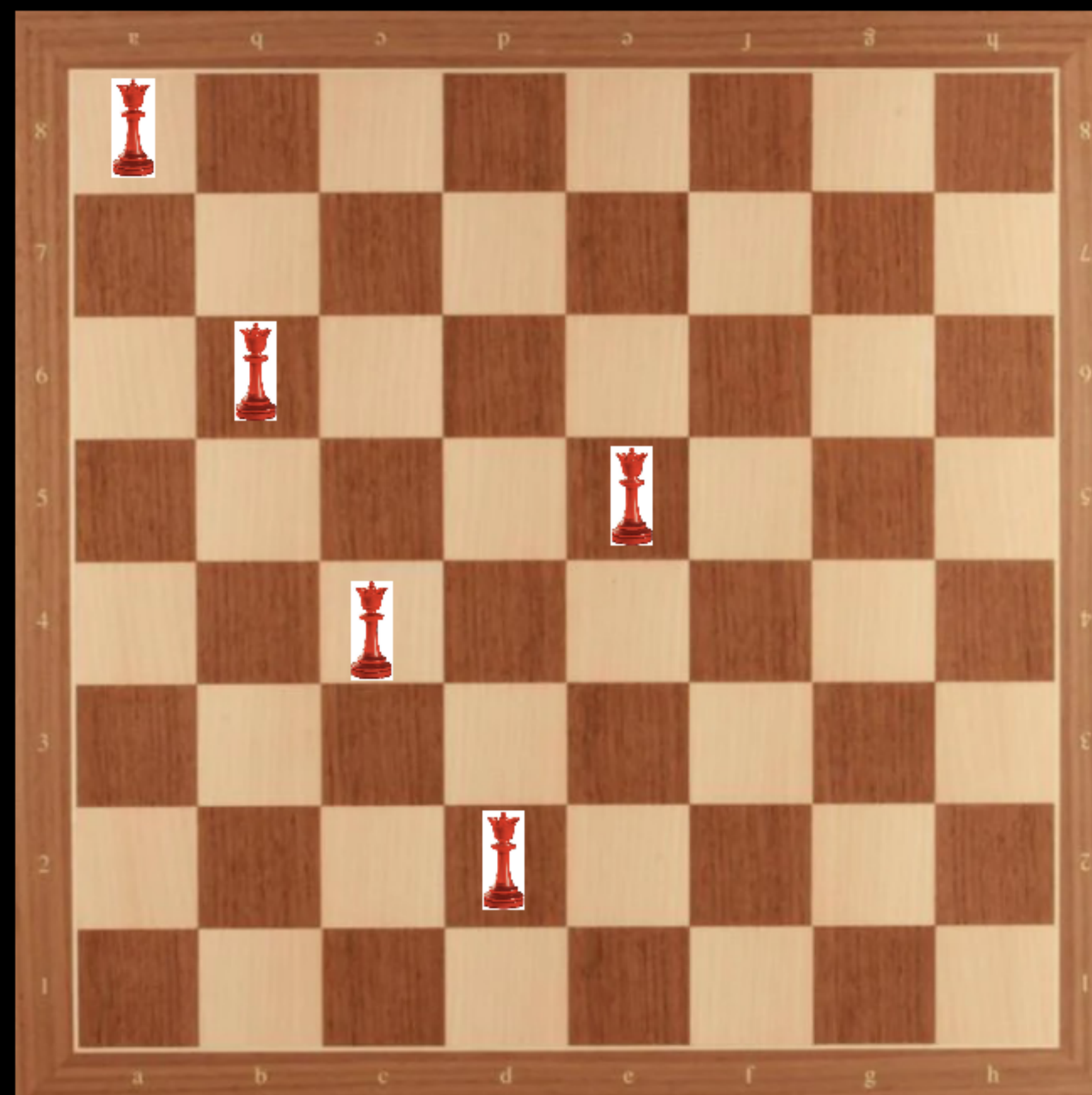
Observation: No two queens can
be on the same row or column.

This reduces the solution space
to $8! = 40,320$ possible
arrangements.

How many arrangements are
solutions?



③



④ PlaceQueen (current Column)

if currentColumn > 8
done — it's solved

else

while (unconsidered squares exist in current Column
and the problem is unsolved)

Determine the next square in the current Column
that is not already under attack by a queen